Evaluating CNN and Logistic Regression Models for Brain Tumor Detection

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ABSTRACT

This project investigates automated brain tumor detection using MRI images from the BR35H 2020 dataset. The objective is to evaluate whether transfer learning with a convolutional neural network can accurately classify tumor vs. non-tumor MRI scans on a relatively small dataset. A MobileNetV2 model pretrained on ImageNet was fine-tuned by unfreezing its final 3 convolutional blocks and retraining the classifier head for binary classification. Data augmentation was applied to reduce overfitting, and model performance was assessed using accuracy, precision, recall, and F1-score on a held-out test set. To provide a meaningful baseline, a Logistic Regression classifier was trained on flattened grayscale images. The fine-tuned MobileNetV2 achieved 99% accuracy and substantially outperformed the baseline model. These results demonstrate that transfer learning is an effective approach for medical image classification when data are limited.

1 INTRODUCTION

Brain tumors are a serious medical condition that often require early detection to improve treatment outcomes. MRI imaging is widely used for diagnosis, but analyzing scans manually is time-consuming and requires expert knowledge. This project addresses the problem of automatically classifying MRI images as either containing a brain tumor or not, using the BR35H 2020 dataset.

The main objective is to evaluate whether transfer learning can produce an accurate classifier with limited training data. Previous work in medical imaging commonly uses pretrained CNN architectures such as EfficientNet(B0-Bn), VGG, ResNet, or MobileNet for tasks like tumor detection, with reported high performance in small datasets. Traditional techniques such as handcrafted feature extraction or classical machine learning can also be used, but they generally perform worse on raw image data.

In this project, a pretrained MobileNetV2 model is fine-tuned on the MRI dataset and compared to a Logistic Regression baseline. The system is evaluated using a train/validation/test split, and results are intended to illustrate the practical effectiveness of transfer learning for machine learning workflows.

2 DATA

This project uses the Br35H :: Brain Tumor Detection 2020 dataset on Kaggle, consisting of MRI images labeled for tumor presence. MRI scans of brain tumors are inherently complex: tumors vary widely in size, shape, and anatomical location, making manual interpretation difficult and time-consuming.

2.1 Data description

- **Number of instances**: 3,060 MRI images
- **Number of attributes/features:** Each image serves as a single instance
- **Image type:** RGB images, resized to 128×128 for modeling
- **Target variable:** Nominal, binary ("yes" for tumor present, "no" for no tumor)
- **Distribution of target variable:** Approximately balanced across the two classes
- **Missing values:** None reported

Figure 1: MRI with tumor present (yes)

Figure 2: MRI with no tumor (no)

2.2 Data understanding

The target variable is binary, indicating whether an MRI scan shows a brain tumor (yes) or not (no).

Each image acts as a single instance, with pixel intensity values implicitly representing features. The relationship between features and the target is that tumor presence alters intensity patterns and texture in the MRI, which the model learns to distinguish. Tumor images vary in size, shape, and location, while non-tumor images capture normal anatomical variation, adding complexity to the classification task.

Current state-of-the-art approaches to tumor classification, or MRI scans as a whole rely on deep learning, particularly CNNs with transfer learning, to achieve high accuracy given the limited size and high complexity of MRI datasets.

2.3 Data preprocessing

Preprocessing was necessary to prepare both the CNN and Logistic Regression models. For the CNN, images were resized to 128×128 pixels, normalized to match the pre-trained MobileNetV2 input distribution, and augmented with random horizontal flips and rotations. These transformations were applied using PyTorch's transforms module.

For the Logistic Regression baseline, images were resized to 32×32 pixels, converted to grayscale, and flattened into 1D feature vectors. These steps were also performed using PyTorch transforms and standard Python, without the use of Scikit-Learn preprocessing tools. Scikit-Learn was used solely for model training and evaluation.

3. MACHINE LEARNING METHODS USED

This project applied two machine learning methods to classify MRI images of brains for tumor presence: a deep learning Convolutional Neural Network (CNN) using transfer learning, and a Logistic Regression (LR) baseline.

The CNN was chosen because convolutional networks are ideal for image classification tasks and can learn spatial patterns directly from pixels. Logistic Regression was included as a simple, interpretable baseline to quantify the advantage of the CNN.

3.1 Brief description of the methods used

CNN with MobileNetV2 (Transfer Learning)

- MobileNetV2 is a lightweight convolutional neural network pre-trained on ImageNet.
- Parameters modified for this project:
  - Last 3 convolutional blocks were unfrozen for fine-tuning.
  - Classifier replaced with a single output neuron for binary classification.
  - Training used Binary Cross-Entropy with logits loss and Adam optimizer with learning rate 1e-4.

Logistic Regression

- Linear model predicting the probability of tumor presence from flattened grayscale images.
- Parameters:
  - Solver: saga
  - Max iterations: 2000
  - Regularization (C=1.0)

3.2 Brief description of the evaluation criteria

The models were evaluated using:

- Accuracy: Overall fraction of correct predictions.
- Precision, Recall, F1-Score: To account for class balance and the importance of avoiding false negatives (missed tumors) and false positives.
- Confusion Matrix: Provides detailed insight into the types of errors made.

These metrics are appropriate for a binary medical classification task, where both correct detection and minimizing false diagnoses are critical.

4. EXPERIMENTS

Experimental Setup

- **CNN Implementation:** PyTorch, MobileNetV2 pre-trained on ImageNet, fine-tuned on Br35H dataset.
- **LR Implementation:** Scikit-Learn Logistic Regression on grayscale flattened images.
- **Data**: 3,060 images split into 70% training, 15% validation, 15% test.
- **Data Augmentation:** Random horizontal flips and rotations applied only to CNN training data
- **Hyperparameters:**

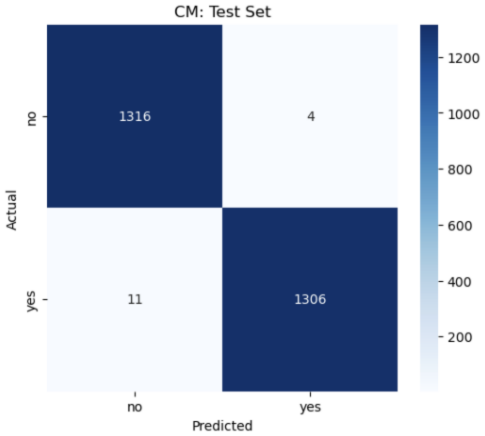
- CNN: 10 epochs, batch size 16, learning rate 1e-4, Adam optimizer
- LR: C=1.0, solver='saga', max_iter=2000

Results

CNN (MobileNetV2):

- Test Accuracy: 0.99
- F1-Score: 0.99

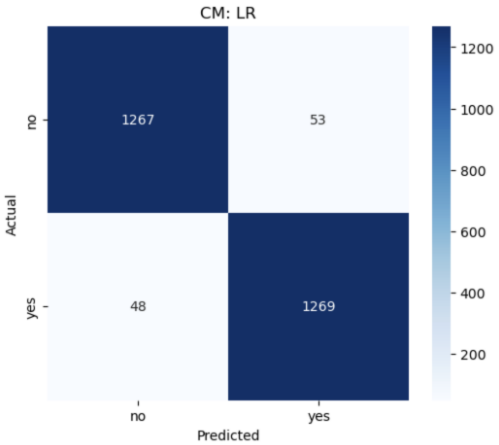
Figure 3: Confusion Matrix for CNN



Logistic Regression:

- Test Accuracy: 0.96
- F1-Score: 0.96

Figure 4: Confusion Matrix for LR



Interpretation of Results

The CNN with transfer learning outperformed Logistic Regression, which is expected because linear models cannot capture the spatial patterns present in MRI images. Tuning the last three MobileNetV2 blocks allowed the CNN to adapt to tumor-specific features, leading to an accuracy and F1-score of 0.99. Overall, the results indicate that transfer learning is far more effective than traditional baseline methods for this type of medical imaging task.

5. VISUALIZATION

Figure 5: Normalized Images for CNN Training

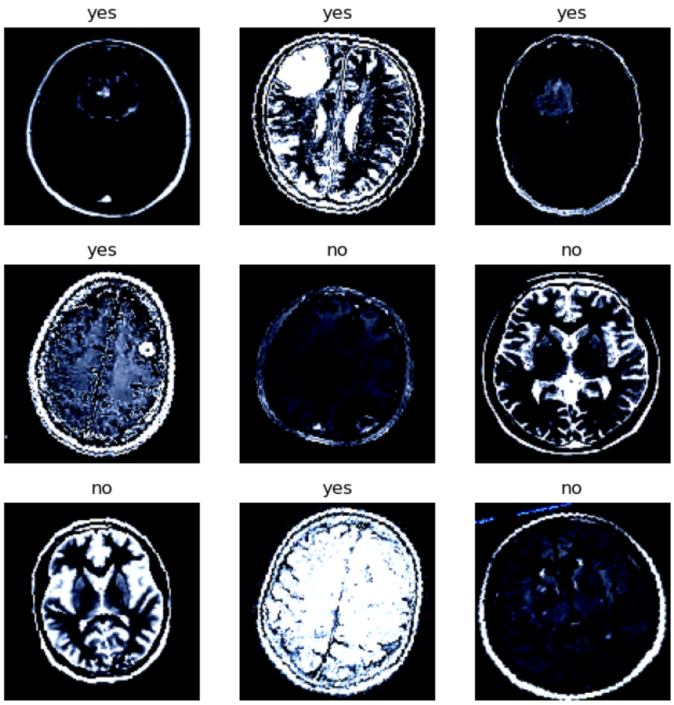


Figure 6: Test Predictions from CNN

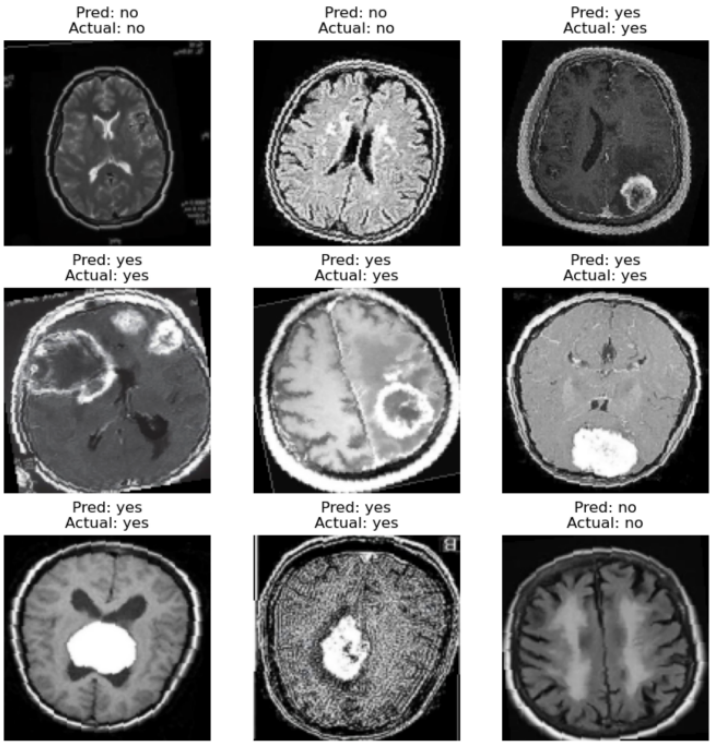
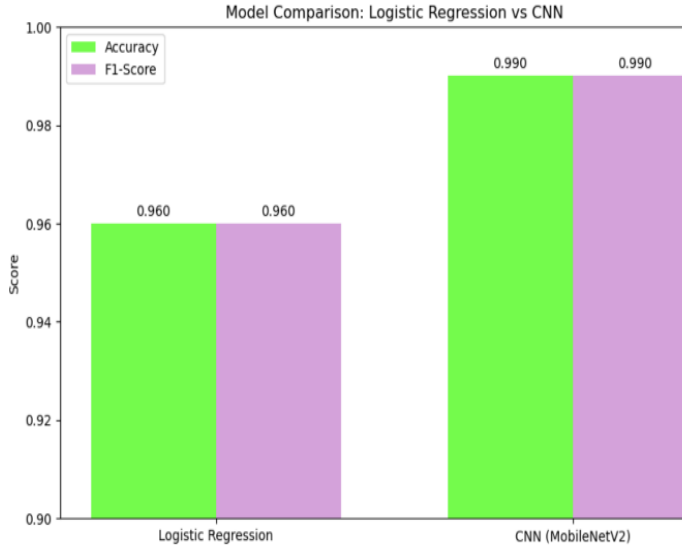


Figure 7: Classification performance between CNN and LR models using accuracy and F1-score.



6. CONCLUSION

This project implemented and compared two approaches for brain tumor detection from MRI images: a Logistic Regression baseline and a transfer-learning CNN based on MobileNetV2. The results showed a clear improvement when using transfer learning, with the CNN achieving near-perfect performance on the test set (≈0.99 accuracy and F1-score). This outcome exceeded initial expectations, as only limited tuning was performed, yet the model was still able to reliably learn the visual patterns.

While simpler models like Logistic Regression are easy to interpret and cheaper computationally, they lack the capacity to capture the complexity of MRI data. In contrast, CNNs (specifically pre-trained models) offer strong performance even with small datasets.

Future work could include exploring more extensive fine-tuning of MobileNetV2, and testing alternative architectures (EfficientNet, VGG),

References

Ahmed Hamada. (2020). Brain Tumor Detection (Br35H) Kaggle. https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection